1. O/N/23/21/Q2

A lorry (truck) is at rest on a straight, horizontal road.

A constant driving force of 5.4×10^4 N acts on the lorry and it accelerates forwards.

(a) Define what is meant by 'acceleration'.

....

....

.... [2]

(b) As the lorry moves forwards, work is done on it by the driving force.

(i) State the equation that defines work done.

Refer to the direction of the force exerted.

....

.... [1]

(ii) Calculate the work done on the lorry by the driving force when the lorry travels a distance of 280 m.

work done = ... J [2]

(iii) At one point on the road, the kinetic energy of the lorry is 3.2×10^6 J and its speed is 12 m / s.

Calculate the mass of the lorry.

mass = ... kg [3]

[Total: 8]

2. M/J/23/22/Q1

Fig. 1.1 shows the speed–time graph for a car travelling on a straight horizontal road.

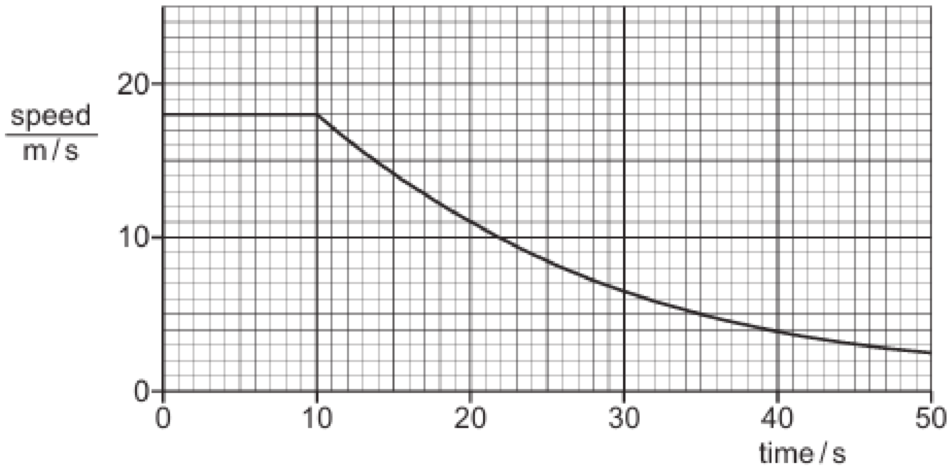


Fig. 1.1

(a) Describe the motion of the car shown in Fig. 1.1.

....

....

....

.... [2]

(b) At time t = 10 s the engine of the car is switched off. The brakes are not applied.

(i) Name **two** forces that act on the car to cause the change in motion after t = 10 s.

1 ....

2 ....

[1]

(ii) Suggest why Fig. 1.1 is a curve after t = 10 s.

....

....

.... [1]

(c) Between t = 10 s and t = 20 s the speed of the car changes from 18 m/s to 11 m/s.

The mass of the car is 1200 kg.

(i) Calculate the change in momentum of the car in this time.

Give the unit of your answer

momentum change = ... unit .... [2]

(ii) Calculate the average resultant force exerted on the car during this time.

average resultant force = ... N [2]

[Total: 8]

3. O/N/22/21/Q7

Fig. 7.1 shows a child sitting on a sledge on a snow-covered hill of constant slope.

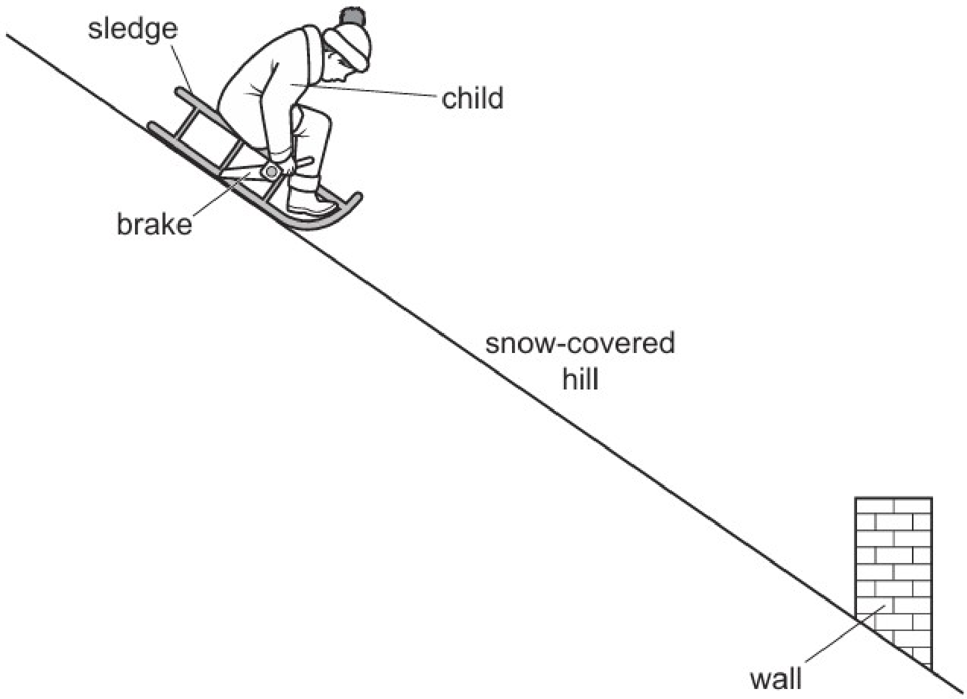


Fig. 7.1 (not to scale)

At time $t = 0$, the child and the sledge begin to move down the hill in a straight line.

When the child sees a wall ahead, he applies the brake.

The child and sledge continue to travel in a straight line until they come to a stop before hitting the wall.

Fig. 7.2 is the speed-time graph for the journey.

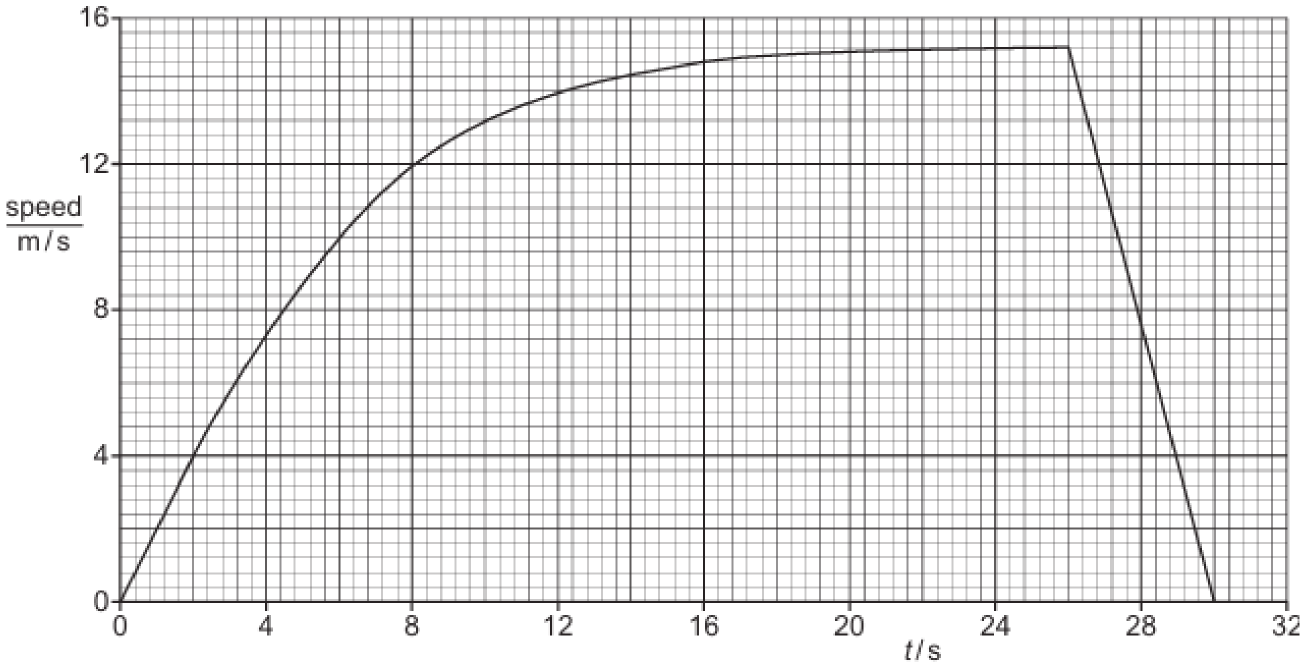


Fig. 7.2

The brake is applied at t = 26 s.

(a) Fig 7.2 shows how the speed of the child and sledge varies over the whole of the journey.

Explain why, between t = 0 and t = 26 s, the speed varies in the way shown by the graph.

....

....

....

....

....

.... [3]

(b) At t = 26 s, the front of the sledge is 35 m from the wall.

Determine the distance between the front of the sledge and the wall when the sledge stops.

distance = ... [3]

(c) At t = 26 s, the child and sledge begin to decelerate.

(i) Determine the size of the deceleration.

deceleration = ... [3]

(ii) The mass of the child is 46 kg and the mass of the sledge is 9.0 kg.

Calculate the resultant force on the child and sledge as they decelerate.

resultant force = ... [2]

(iii) State the energy transfer that is taking place as the child and sledge decelerate.

....

....

.... [2]

(d) At t = 26 s, when the brake is first applied, the child jerks forwards on the sledge.

Explain why.

....

....

.... [2]

[Total: 15]

4. O/N/22/22/Q1

A train travels along a straight horizontal track. At time $t = 0$, the train passes through station P at constant speed without stopping.

The driver applies the brakes 70 s before reaching station Q. The train decelerates.

Fig. 1.1 is the speed–time graph for the train from $t = 0$ until it stops at station Q.

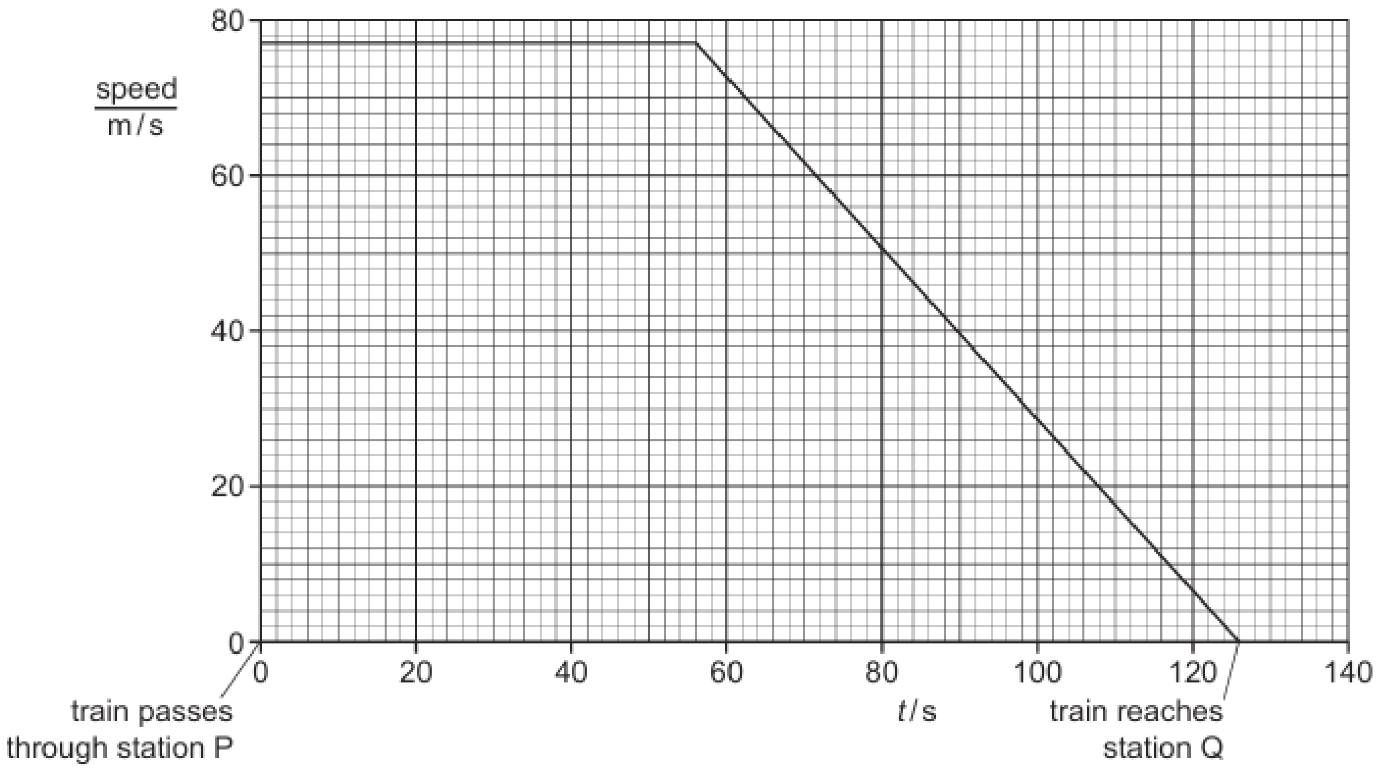


Fig. 1.1

(a) Using Fig. 1.1, determine the distance between station P and station Q.

distance = ... [3]

(b) The mass of the train is 3.8×10^5 kg.

(i) Determine the deceleration of the train in the 70 s before it stops at station Q.

deceleration = ... [2]

(ii) Calculate the resultant force on the train as it decelerates.

resultant force = ... [2]

[Total: 7]

5. M/J/22/22/Q7

Fig. 7.1 shows the speed–time graph for a car travelling on a straight horizontal road.

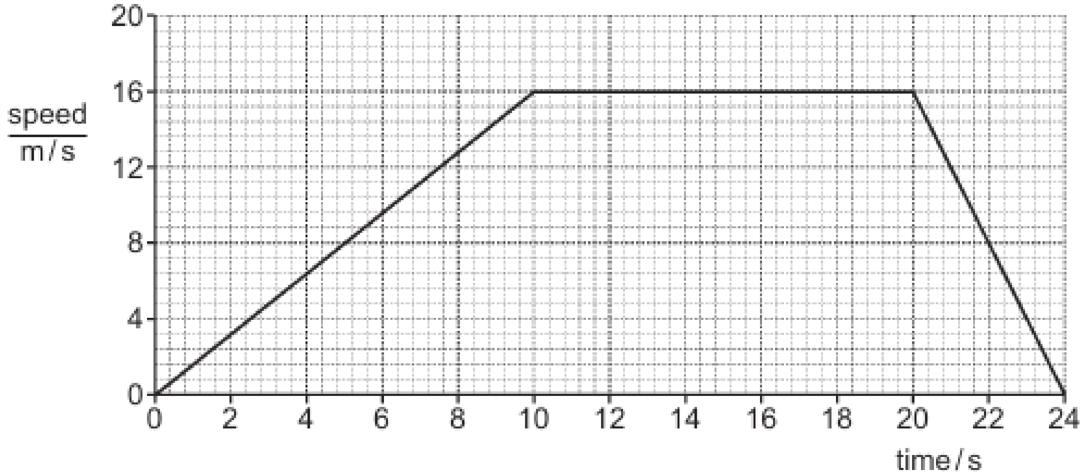


Fig. 7.1

(a) **(i)** Describe the motion of the car.

....

....

....

....

.... [3]

(ii) Using Fig. 7.1, calculate the distance travelled by the car during the 24 s of its motion.

Show your working.

distance = ... [3]

(iii) Calculate the average speed of the car during its motion.

average speed = ... [2]

(iv) A second car travels at a steady speed. It travels the same distance as the first car in the 24 s of the journey.

On Fig. 7.1, draw the speed–time graph for the second car. [2]

(b) The thinking distance is the distance travelled by a car between the time that a hazard is seen and the time that the brakes are applied.

The braking distance is the distance travelled while the car slows down to rest.

Table 7.1 shows the thinking and braking distances for an alert driver when the car travels at different speeds.

Table 7.1

$\frac{\text{speed}}{\text{km/h}}$	thinking distance / m	braking distance / m
20	9	2
40	18	9
60		20
80	36	36
100	45	56

(i) Complete Table 7.1. [1]

(ii) The time it takes for the driver to react to the hazard is constant at different speeds.

Explain how the table shows this.

....

....

....

.... [2]

(iii) State what happens to the thinking distance and the braking distance when the driver is tired.

thinking distance ....

braking distance ....
[2]

[Total: 15]

6. M/J/22/21/Q7(c)

Fig. 7.1 shows a toy helicopter. It can hover and travel through the air.

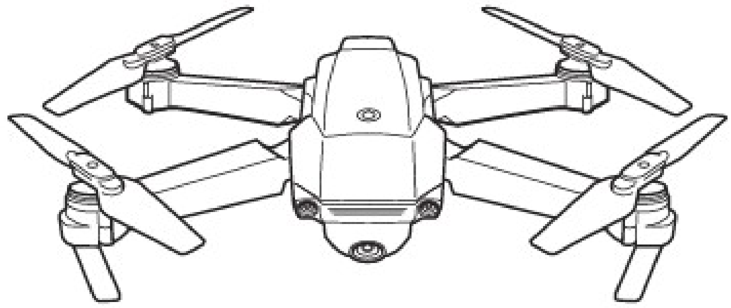


Fig. 7.1

A student flies the toy helicopter on a journey from A to B to C to D at a constant height.

Fig. 7.2 is a scale drawing of the path of the helicopter, viewed from above.

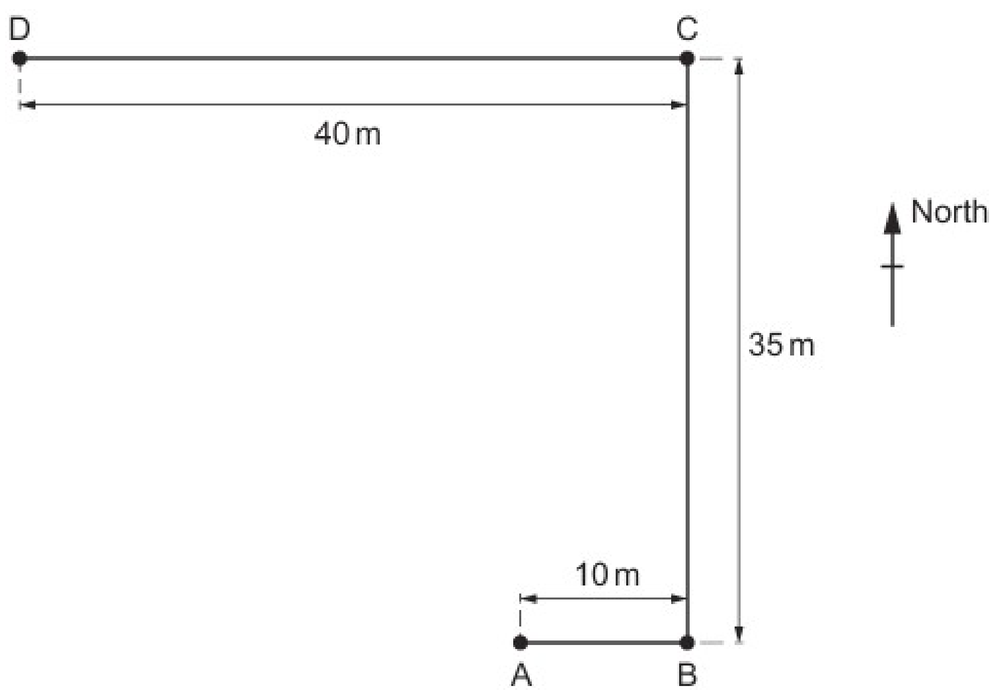


Fig. 7.2 (to scale)

(c) When the toy helicopter hovers at D, its motor fails and it falls. It reaches terminal velocity as it falls.

Explain, in terms of the forces and acceleration, what happens as the helicopter falls and reaches terminal velocity.

....

....

....

....

....

....

....

.... [5]

7. O/N/21/21/Q2

Fig. 2.1 shows a man of mass 80 kg standing in a lift (elevator).

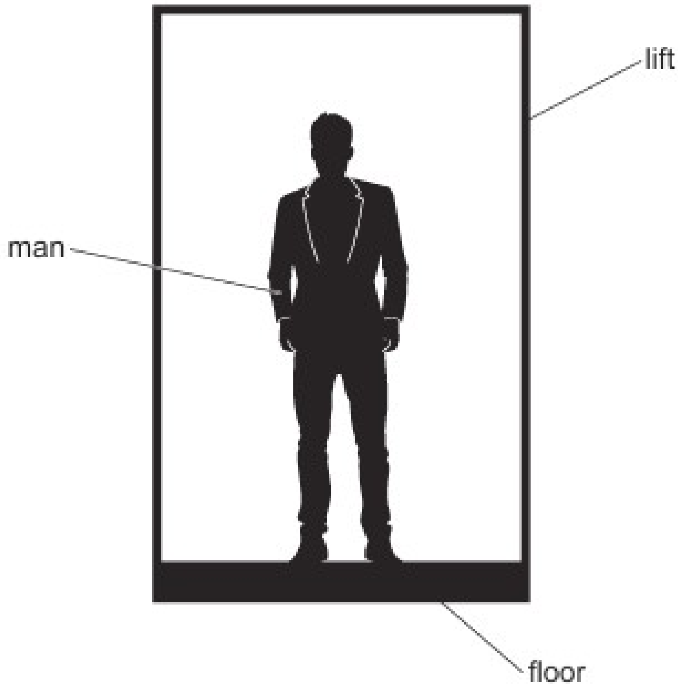


Fig. 2.1

The gravitational field strength g is 10 N / kg.

(a) Calculate the weight of the man.

weight = ... [1]

(b) The lift accelerates upwards uniformly at 0.50 m / s^2.

Calculate:

(i) the resultant upward force on the man

resultant force = ... [2]

(ii) the force exerted on the man by the floor of the lift.

force = ... [1]

(c) The lift accelerates upwards uniformly from rest for 4.0 s and then decelerates uniformly to rest in 4.0 s. The total distance travelled is 8.0 m.

On Fig. 2.2, sketch the distance-time graph for this journey.

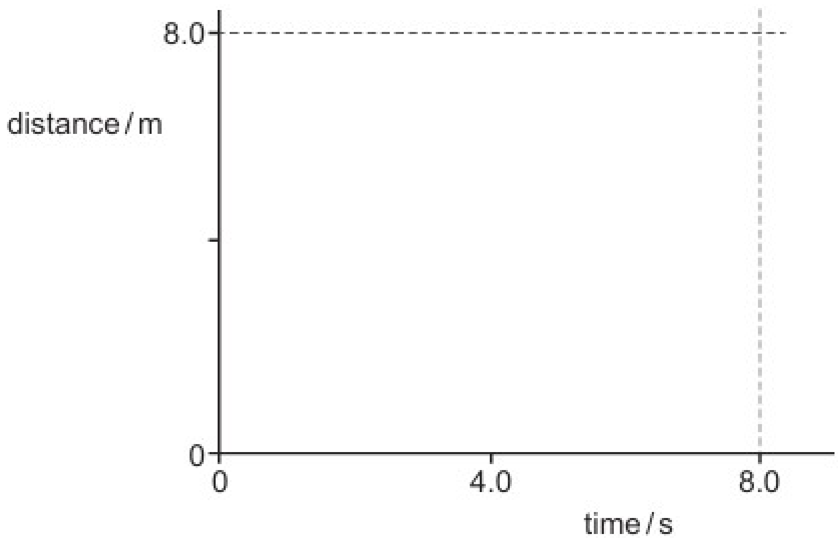


Fig. 2.2

[3]

[Total: 7]

8. O/N/21/22/Q2

Force is a vector quantity.

(a) State what is meant by a *vector*.

....

.... [1]

(b) A swimmer reaches the end wall of a swimming pool and turns around under the water.

Fig. 2.1 shows the swimmer immediately after turning around.

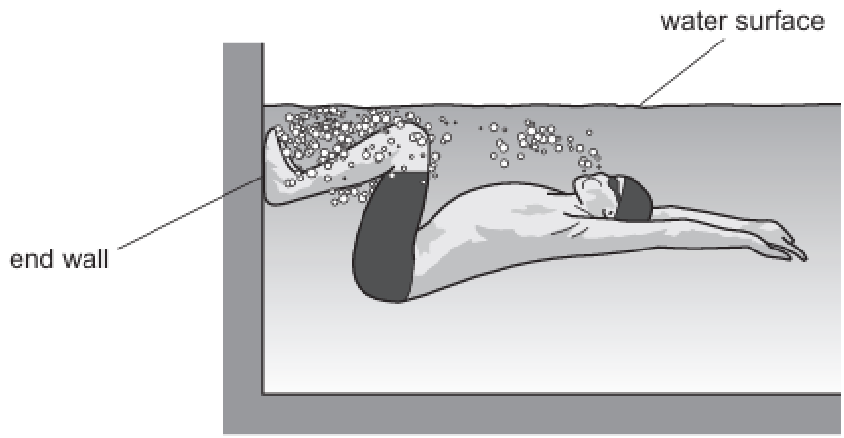


Fig. 2.1

(i) The swimmer pushes against the end wall of the pool with his legs.

Explain, in terms of Newton's third law, why the swimmer accelerates away from the end wall.

....

....

....

.... [3]

(ii) While swimming, there is a constant forward force on the swimmer. His speed increases until eventually he reaches a constant speed.

Explain why he reaches a constant speed.

....

....

....

.... [3]

[Total: 7]

9. M/J/21/21/Q8

Fig. 8.1 shows a stationary horse and its rider, about to jump over two fences.

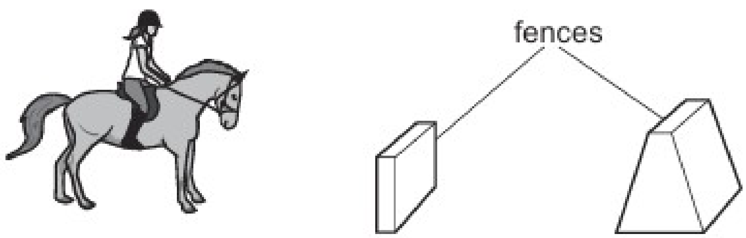


Fig. 8.1

(a) Fig. 8.2 shows a side view of the horse.

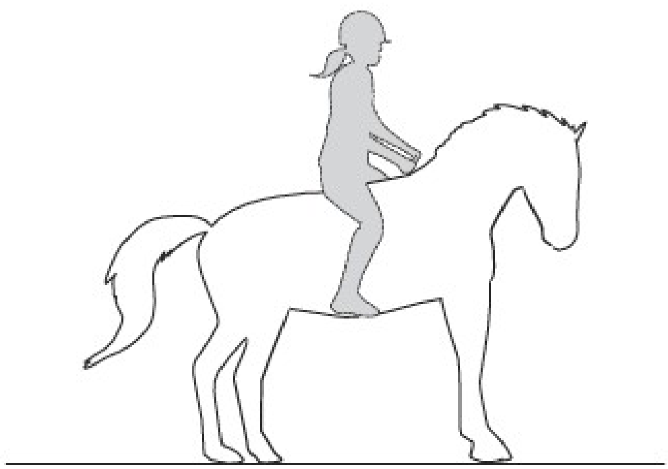


Fig. 8.2

(i) On Fig. 8.2, draw and label the forces acting on the horse.

Include the force that the rider exerts on the horse. Label this force *F*. [3]

(ii) Explain how Newton's third law applies to force *F*.

....

....

.... [2]

10. O/N/20/21/Q1

Fig. 1.1 is the distance–time graph for a skydiver who jumps from a balloon at time t = 0.

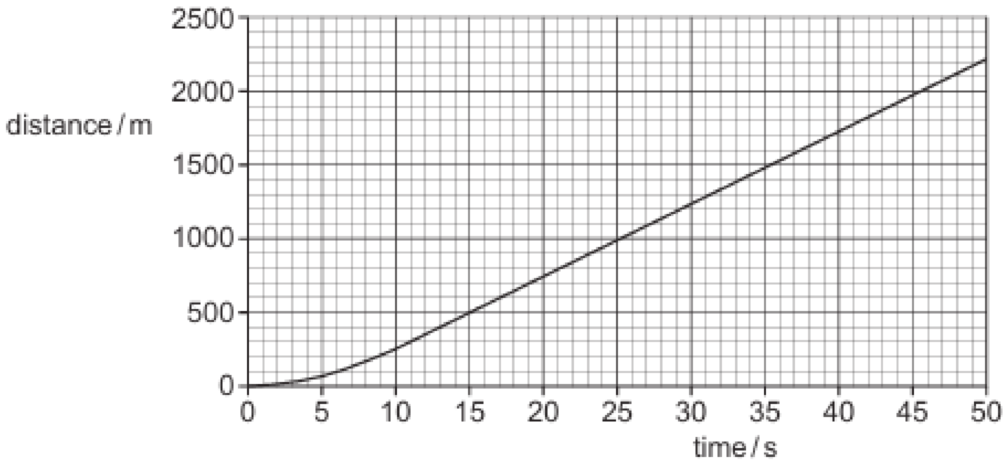


Fig. 1.1

(a) The first part of the graph shows the motion of the skydiver from when he jumps until he reaches terminal velocity.

(i) Describe the motion of the skydiver between t = 0 and t = 20 s.

....

....

.... [2]

(ii) Explain the motion of the skydiver between t = 0 and t = 20 s in terms of the forces acting on him.

....

....

....

.... [3]

(b) Using Fig. 1.1, determine the terminal velocity of the skydiver.

On Fig. 1.1, indicate any values used for your calculation.

terminal velocity = ... [3]

[Total: 8]

11. O/N/20/21/Q2

Fig. 2.1 shows a satellite moving at a constant speed in a circular orbit around the Earth.

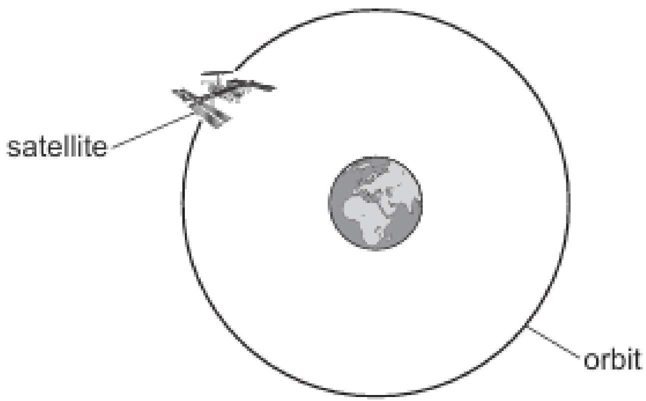


Fig. 2.1 (not to scale)

Speed is a scalar quantity but velocity is a vector quantity.

(a) State how a scalar quantity differs from a vector quantity.

....

.... [1]

(b) Underline every vector quantity in the list.

distance **displacement** **force** **length** **mass** **time**

[1]

(c) There is a resultant force acting on the satellite in Fig. 2.1.

(i) Explain how the motion of the satellite shows that a resultant force is acting on it.

....

....

.... [2]

(ii) State the cause of this force.

....

.... [1]

[Total: 5]

12. O/N/20/22/Q7

A bus leaves a bus-stop at time $t = 0$ and travels along a horizontal road until it reaches a second bus-stop. Fig. 7.1 is the distance-time graph for the bus between $t = 0$ and $t = 60\,\text{s}$.

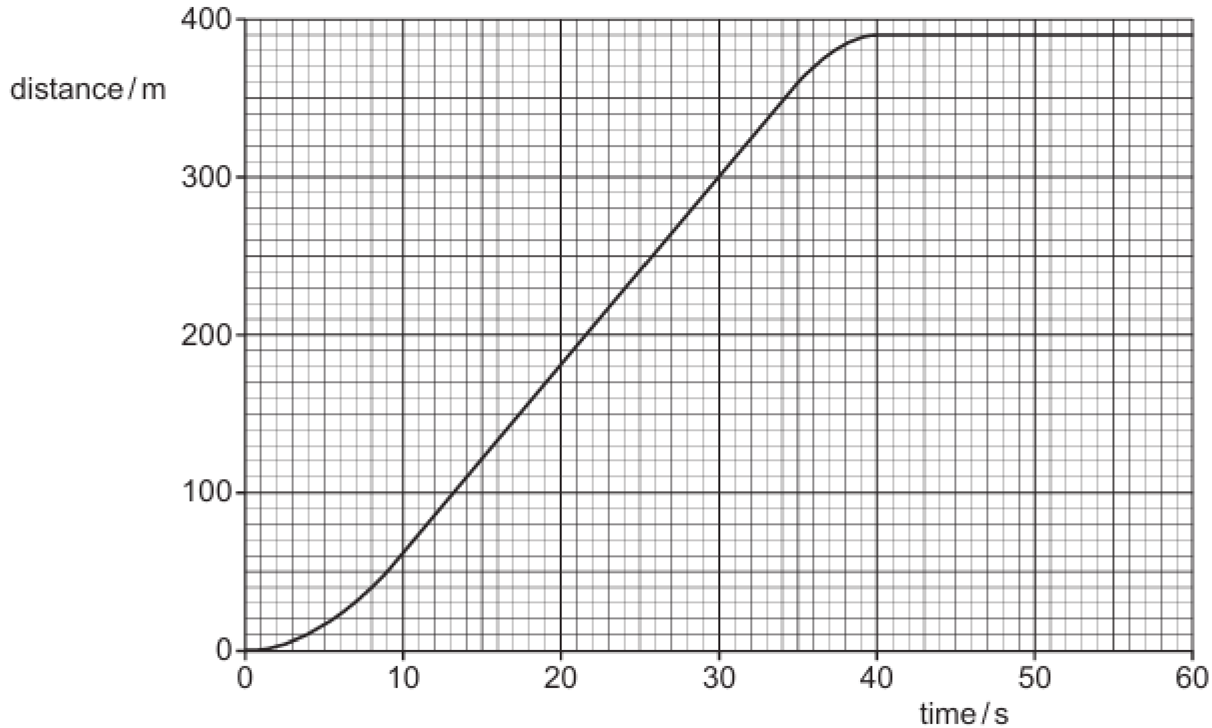


Fig. 7.1

The road on which the bus is travelling is straight except for a short, curved section. The bus travels around this circular curve between $t = 21\,\text{s}$ and $t = 24\,\text{s}$.

(a) Describe how the motion of the bus between $t = 0$ and $t = 10\,\text{s}$ differs from its motion between $t = 35\,\text{s}$ and $t = 40\,\text{s}$.

....

....

....

....

.... [3]

(b) Determine:

(i) the maximum speed of the bus during these 60 s

maximum speed = ... [3]

(ii) the average speed of the bus between leaving the first bus-stop and arriving at the second bus-stop.

average speed = ... [2]

(c) **(i)** State how *velocity* differs from *speed*.

....

.... [1]

(ii) There are **three** periods during the 60 s when there is a non-zero resultant force acting on the bus.

Complete the statements to indicate these three time periods and state the direction of the resultant force in that period.

1. Between t = ... and t = ... the direction of the resultant force is

....

2. Between t = ... and t = ... the direction of the resultant force is

....

3. Between t = ... and t = ... the direction of the resultant force is

....

[4]

(d) During the journey, the air resistance acting on the bus varies.

(i) State why the air resistance changes during the journey.

....

.... [1]

(ii) On Fig. 7.1, mark and label with an M a time when the air resistance is a maximum value. [1]

13. M/J/20/22/Q2

Fig. 2.1 shows the thinking distance and the braking distance for a car driven at 100 km/h.

The car has old, smooth tyres.

Fig. 2.1

(a) Calculate the total stopping distance for the car.

stopping distance = ... [1]

(b) The car is now fitted with new tyres.

At a speed greater than 100 km/h, the total stopping distance is the same as in **(a)**.

(i) State and explain the effect that the increase in speed and the use of new tyres have on the thinking distance.

effect ....

explanation ....

....

.... [2]

(ii) State and explain the effect that the increase in speed and the new tyres have on the braking distance.

effect ....

explanation ....

....

.... [2]

[Total: 5]

14. M/J/20/21/Q1

Fig. 1.1 shows the thinking distance and the braking distance for a car being driven along a dry road and along a wet road at the same speed.

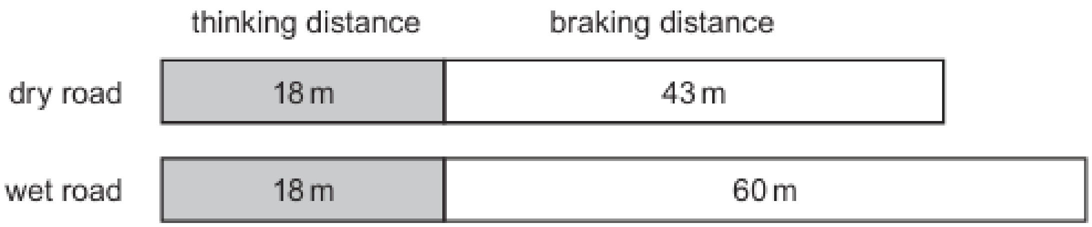


Fig. 1.1

(a) Calculate the total stopping distance for the car on the wet road.

distance = ... [1]

(b) Complete the sentence.

The thinking distance is the distance travelled between seeing a hazard and ....

.... [1]

(c) **(i)** Suggest why the thinking distance is the same on both roads.

....

....

.... [1]

(ii) Explain why the braking distance is larger when the road is wet.

....

....

....

.... [2]

[Total: 5]

15. M/J/19/22/Q9(b)

Fig. 9.1 shows a map of the route taken by a car as it travels from town A to town B.

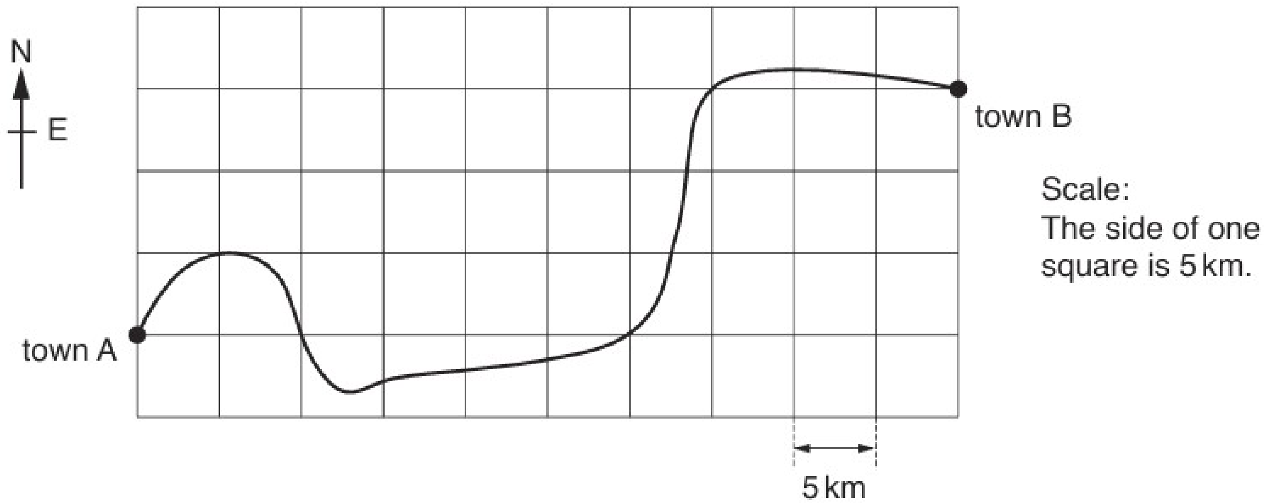


Fig. 9.1

(b) At town B the car makes an emergency stop.

The total stopping distance of the car is the thinking distance added to the braking distance.

(i) Describe what is meant by the *thinking distance*.

....

.... [1]

(ii) The brakes on the car work badly. This increases the braking distance.

State **two** other factors that increase the braking distance.

1. ....

....

2. ....

....

[2]

16. M/J/19/21/Q9

Fig. 9.1 shows a parachutist falling vertically towards the ground.

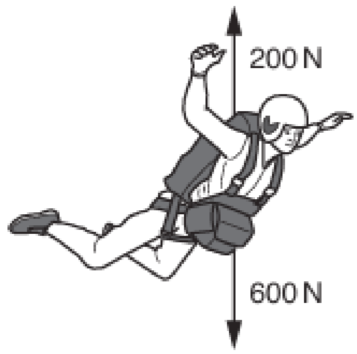


Fig. 9.1

The mass of the parachutist is 60 kg and his weight is 600 N.

(a) State what is meant by:

(i) *mass*

....

.... [1]

(ii) *weight.*

....

.... [1]

(b) Calculate the acceleration of the parachutist when the air resistance is 200 N.

acceleration = ... [3]

(c) The parachutist falls from rest at time $t = 0$.

He accelerates non-uniformly until he reaches a terminal velocity of 44 m/s at $t = 30$ s.

At $t = 40$ s the parachute opens. He decelerates for 4.0 s, reaching a smaller terminal velocity of 5.0 m/s.

He then falls at this speed until he lands on the ground at $t = 200$ s.

(i) On Fig. 9.2, draw the speed–time graph for the parachutist.

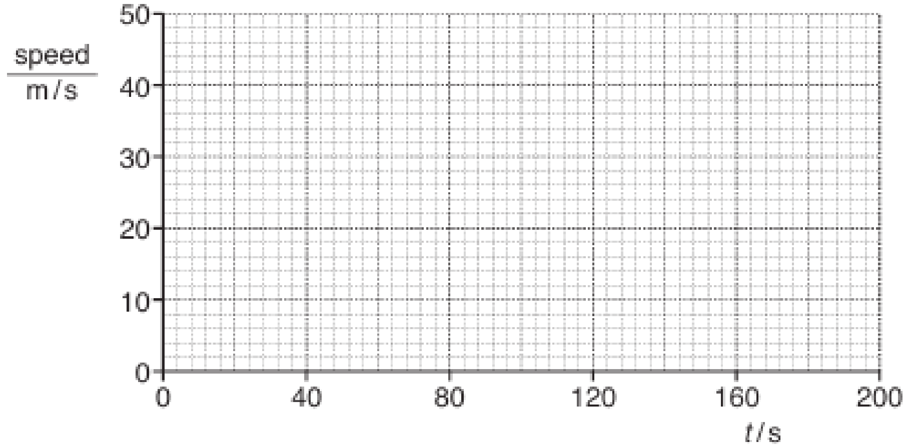


Fig. 9.2

[4]

(ii) Calculate the average deceleration of the parachutist between $t = 40$ s and $t = 44$ s.

deceleration = ... [2]

(d) Explain, in terms of the forces involved, why:

(i) the parachutist reaches a terminal velocity at $t = 30$ s

....

.... [1]

(ii) the parachutist decelerates when the parachute opens

....

....

.... [2]

(iii) the second terminal velocity, when the parachute is open, is smaller than the first.

....

.... [1]

17. O/N/18/21/Q1

A girl of mass 35 kg, on a bicycle, accelerates from rest and travels down a slope in a straight line. The girl does not use the pedals.

Fig. 1.1 shows that the gradient of the slope is constant.

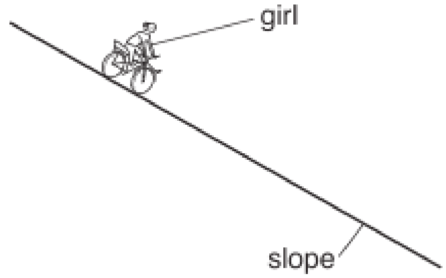


Fig. 1.1

(a) Calculate the resultant force on the girl when she is accelerating at 2.6 m/s^2.

resultant force = ... [2]

(b) At first, her acceleration is constant. At time t_1, her acceleration starts to decrease gradually until she is travelling at a constant speed in a straight line.

(i) On Fig. 1.2, sketch a speed-time graph for the girl from when she starts moving until she is travelling at a constant speed. [2]

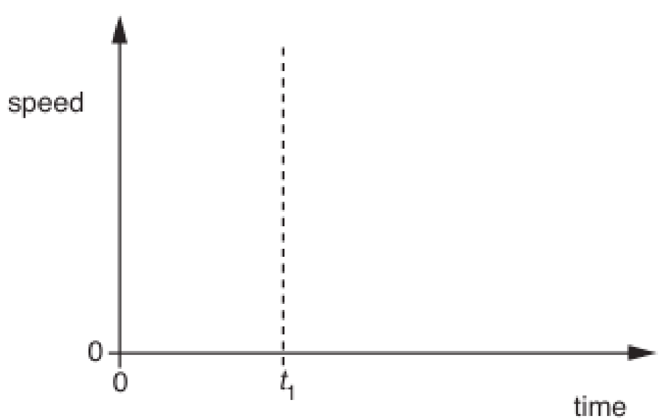


Fig. 1.2

(ii) State how the distance travelled is found from a speed-time graph.

....

.... [1]

18. M/J/18/21/Q2

Fig. 2.1 shows a man pushing a heavy box with a force *P*. A frictional force *F* acts in a horizontal direction.

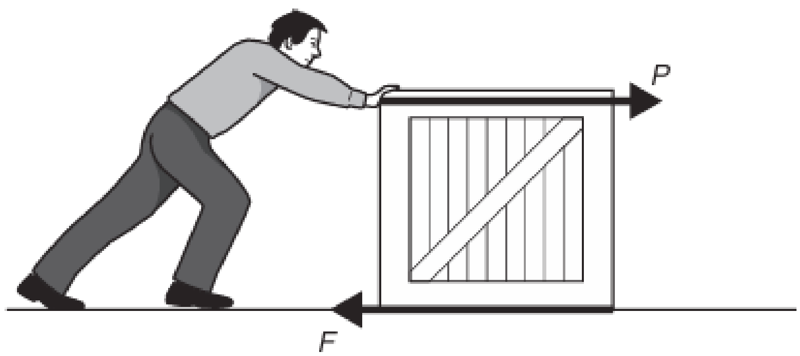


Fig. 2.1

(a) The forces on the box are *balanced* and the box is stationary.

(i) State what is meant by *balanced forces*.

....

....[1]

(ii) The frictional force in Fig. 2.1 does not produce any heating effect.

State what must happen for the frictional force to produce heating.

....

....[1]

(iii) Apart from being stationary, describe one other possible state of motion of the box when the forces are balanced.

....

....[1]

(b) When $P = 100\,\text{N}$ and $F = 85\,\text{N}$, the box accelerates. The mass of the box is 25 kg.

Calculate the acceleration of the box.

acceleration = ...[3]

19. M/J/17/21/Q1

Fig. 1.1 shows the directions of four forces acting on a racing car as it travels in a horizontal straight line.

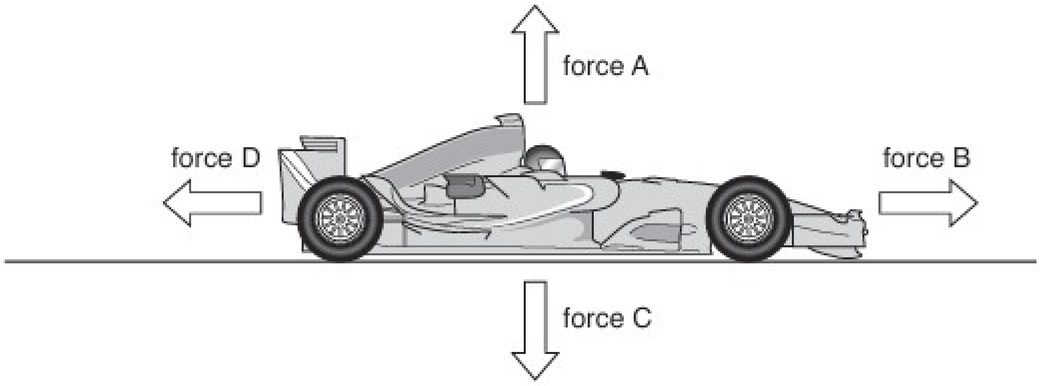


Fig. 1.1

(a) Draw a line from each box on the left to the correct description of each force.

force A	driving force
force B	contact or normal reaction force
force C	air resistance and friction
force D	force of gravity

[1]

(b) The table shows the sizes of the forces acting on the car at one time.

force A / N	force B / N	force C / N	force D / N
8000	1000	8000	600

The gravitational field strength g is 10 N / kg.

Calculate

(i) the mass of the car,

mass = ...[1]

(ii) the resultant force on the car,

resultant force = ...[1]

(iii) the acceleration of the car.

acceleration = ...[2]

(c) At another time, the car is travelling at speed u. It then accelerates for 5.0 s with an acceleration of $1.6\,m/s^2$, and reaches a speed of 20 m/s.

Calculate the value of u.

u = ...[2]

20. M/J/16/21/Q12

The overall stopping distance of a cyclist is made up of two parts, as shown in Fig. 12.1:

- the distance the cyclist travels during the reaction time of the cyclist (the thinking distance)
- the distance the cyclist travels after the brakes are applied (the braking distance)

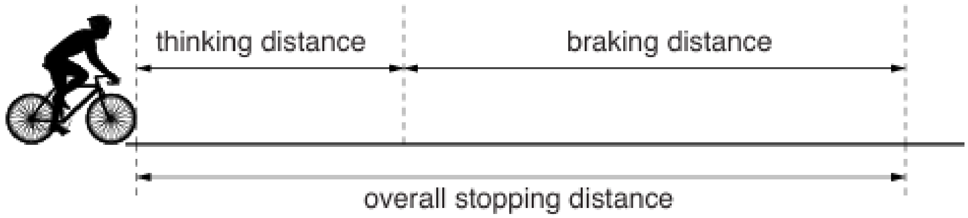


Fig. 12.1 (not to scale)

(a) State the energy change that occurs during braking.

....

....[2]

(b) A ball rolls in front of a cyclist at time $t = 0$ and the cyclist brakes and comes to rest.

Fig. 12.2 shows the speed-time graph for the cyclist.

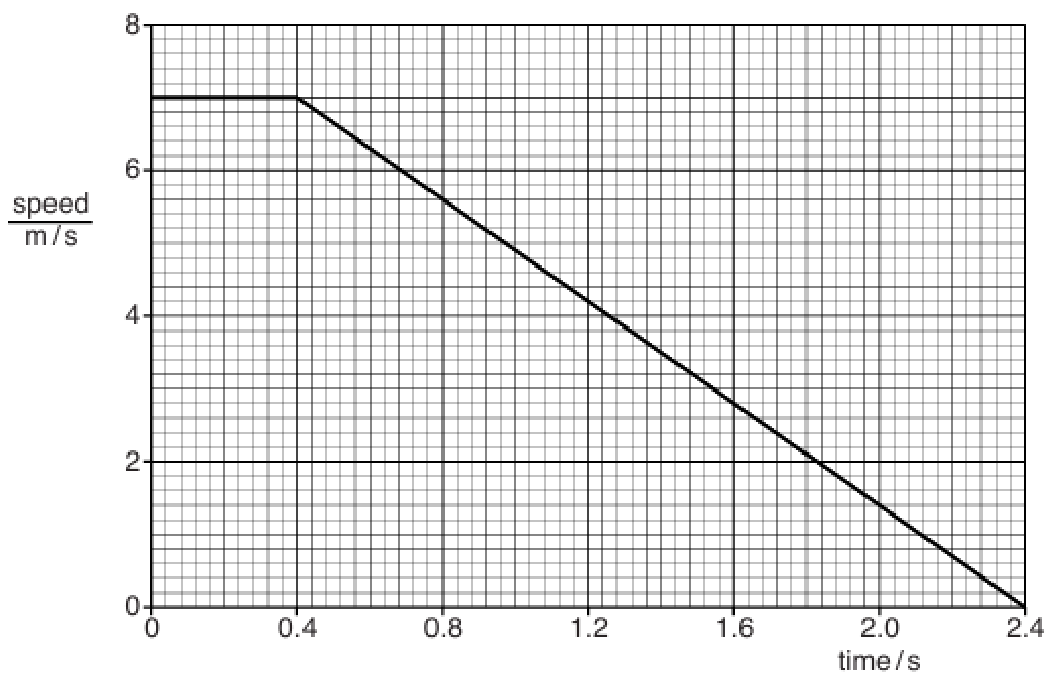


Fig. 12.2

(i) Using Fig. 12.2, determine the reaction time of the cyclist.

reaction time = ...[1]

(ii) Using Fig. 12.2, calculate the thinking distance.

thinking distance = ...[2]

(iii) State how the braking distance is found using Fig. 12.2.

....

....[1]

(iv) On another occasion, the same cyclist travels at an initial speed of 5.0 m/s. A ball rolls in front of the cycle at time $t = 0$.

The cyclist has the same reaction time and the deceleration of the cycle is the same as in Fig. 12.2.

On Fig. 12.2 draw the new speed-time graph for the cyclist. [2]

(v) The braking distance is longer when the cyclist stops on a wet road.

Explain why.

....

....

....

....

....

....[3]

21. M./J/14/22/Q1

Fig. 1.1 shows a lorry accelerating in a straight line along a horizontal road.

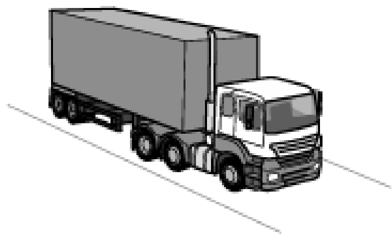


Fig. 1.1

(a) The driving force on the lorry in the forward direction is D and the total backward force on the lorry is B.

(i) State and explain whether D or B is the larger force.

....

.... [1]

(ii) Suggest one possible cause of the backward force B.

.... [1]

(b) The weight of the lorry is 300 000 N.

The gravitational field strength g is 10 N / kg.

(i) Calculate the mass of the lorry.

mass = ... [1]

(ii) The resultant force on the lorry is 15 000 N. Calculate the acceleration of the lorry.

acceleration = ... [2]

(c) Later, the lorry turns a corner at constant speed.

Explain why the lorry accelerates even though the speed is constant.

....

.... [1]

22. O/N/11/22/Q2

Fig. 2.1 shows a skier of mass 85 kg skiing down a very steep slope.

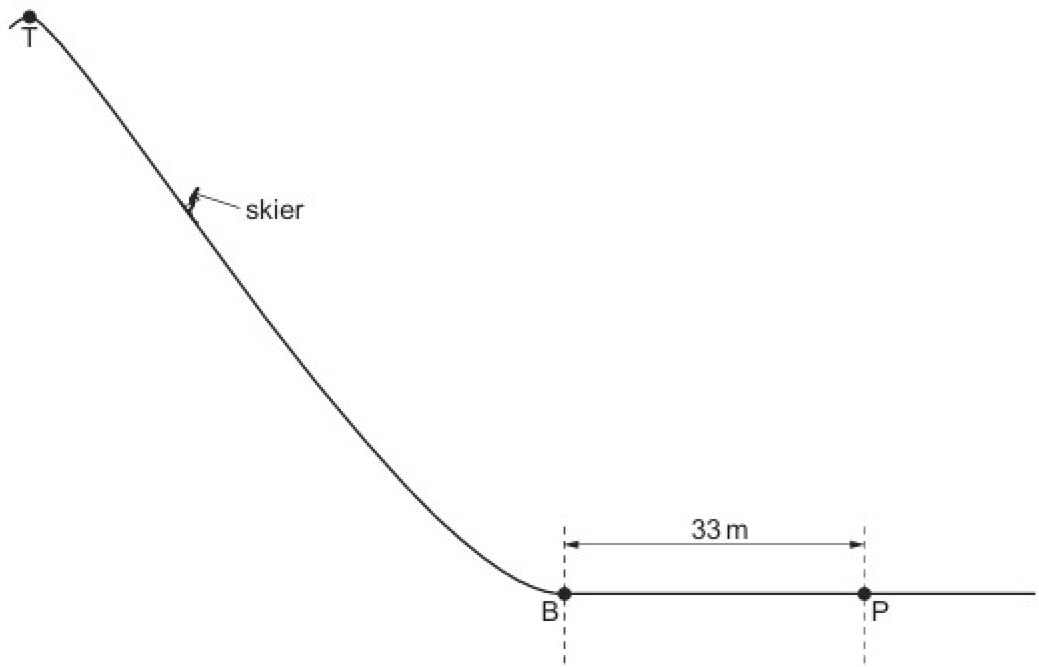


Fig. 2.1 (not to scale)

The skier starts from rest at the top T of the slope. The force of gravity accelerates him down the slope.
When he reaches the bottom B of the slope, his kinetic energy is 5.5×10^4 J.

(a) The gravitational field strength is 10 N / kg. Calculate

(i) the weight of the skier,

weight = ...[1]

(ii) the minimum possible difference in height between T and B.

height difference = ...[2]

(b) At B, the skier digs his skis into the snow and stops at the point P after travelling 33 m horizontally.

Calculate the average horizontal force that acts on the skier between B and P.

force = ...[2]

23. M/J/15/21/Q9

Fig. 9.1 shows a satellite in orbit around the Earth.

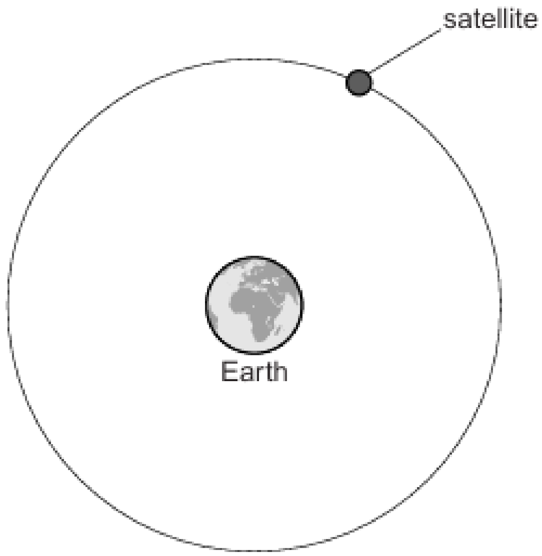


Fig. 9.1 (not to scale)

(a) The satellite travels at a constant speed in a circular orbit.

(i) Underline the quantities in the list below that are scalars.

acceleration force mass speed velocity [2]

(ii) The velocity of the satellite changes, but its speed is constant.

1. State what is meant by *velocity*. ....

....

....[1]

2. Explain why the velocity changes. ....

....

....[1]

(iii) Explain what makes this satellite move in an orbit that is circular.

....

....

....[2]

(b) The satellite is placed into orbit by a rocket. Fig. 9.2 shows the rocket as it takes off.

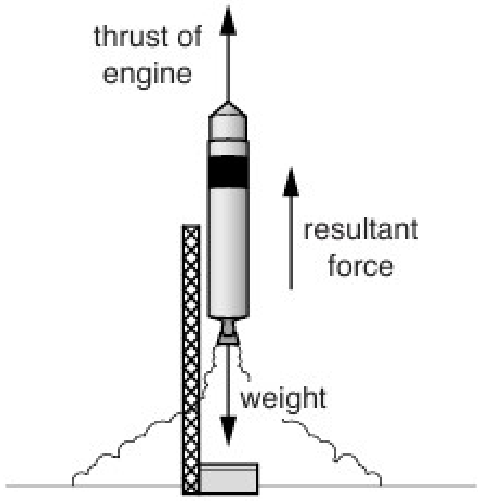


Fig. 9.2

The rocket and fuel have a total mass of 40 000 kg and a total weight of 400 000 N. The resultant force acting upwards on the rocket is 50 000 N.

(i) Calculate the thrust produced by the rocket engine.

thrust = ...[1]

(ii) Calculate the acceleration of the rocket.

acceleration = ...[2]

24. M/J/12/22/Q2

(a) State what is meant by *friction*.

....

....[2]

(b) Fig. 2.1 shows two horizontal forces that act on a car. Force B is caused by air resistance and friction.

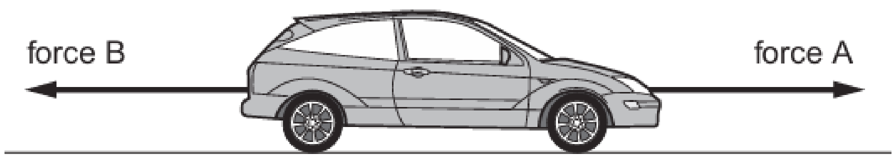


Fig. 2.1

The car is travelling along a straight level road.

(ii) The mass of the car is 800 kg.
Force A increases to 5000 N. This causes the car to accelerate initially at 1.5 m/s^2.
Calculate the size of force B.

force B = ...[3]

(iii) Force A remains constant at 5000 N. Explain why the acceleration decreases as the car travels along the level road.

....

....[2]

25. O/N/11/21/Q9

Fig. 9.1 is the speed-time graph for a rocket from the moment that the fuel starts to burn at time $t = 0$.

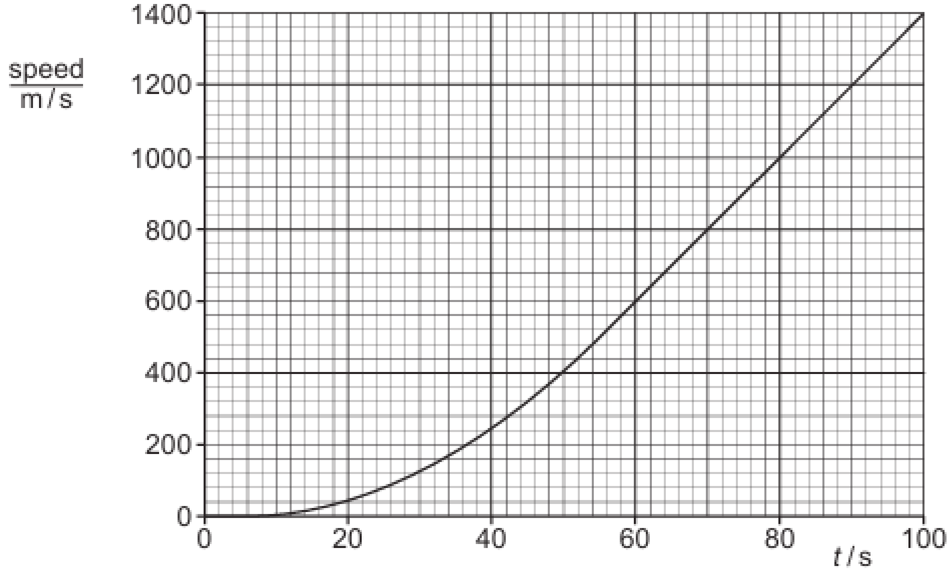


Fig. 9.1

(a) State the principal energy changes taking place as the rocket accelerates upwards.

....

....

....

....[4]

(b) **(i)** State the size of the acceleration of the rocket at $t = 0$.

....[1]

(ii) State what happens to the acceleration of the rocket between $t = 5$ s and $t = 80$ s.

....

....[2]

(iii) Calculate the acceleration of the rocket at $t = 80$ s.

acceleration = ...[2]

(iv) The total mass of the rocket at $t = 80$ s is 1.6×10^6 kg. Calculate the resultant force on the rocket at this time.

resultant force = ...[2]

(v) The total weight of the rocket at $t = 80\,\text{s}$ is $1.6 \times 10^7\,\text{N}$. Calculate the upward force on the rocket at this time, caused by the burning fuel.

upward force = ...[1]

(c) As the rocket burns fuel, it ejects hot gas downwards.

(i) State *Newton's third law of motion.*

....

....[1]

(ii) Explain how Newton's third law of motion applies to the upward force on the rocket and to the force on the hot gas.

....

....

....[2]

26. M/J/11/21/Q2

Fig. 2.1 shows a sky-diver falling vertically downwards at terminal velocity.

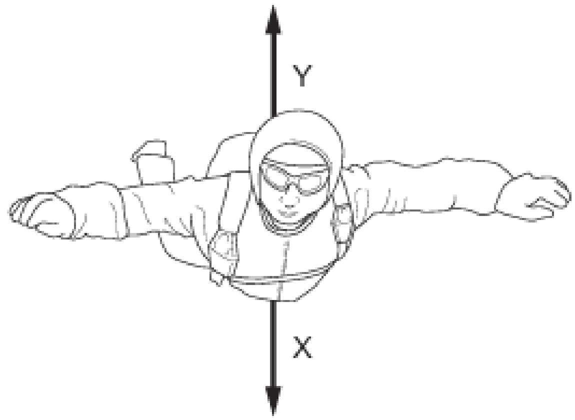


Fig. 2.1

Arrows X and Y show the two main forces acting on the sky-diver.

(a) **(i)** State the name of force X and the name of force Y.

X ....

Y .... [1]

(ii) Explain why force Y acts upwards.

....

.... [1]

(b) When the sky-diver first started to fall, forces X and Y were unbalanced.

(i) Describe and explain the effect of the unbalanced forces on the motion of the sky-diver.

....

....

.... [2]

(ii) State what happened to the size of force X and the size of force Y as the sky-diver fell and reached terminal velocity.

....

....

.... [2]

27. O/N/10/21/Q2

Fig. 2.1 shows the planet Venus orbiting the Sun in a circular orbit at constant speed.

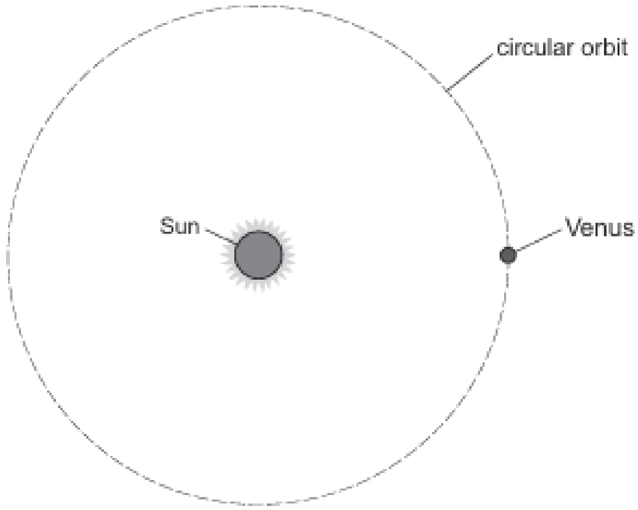


Fig. 2.1 (not to scale)

A force acting on Venus keeps it moving in a circle.

(a) State the name of this force and the object that causes it.

....

.... [2]

(b) **(i)** On Fig. 2.1, draw an arrow through Venus that represents the direction of its acceleration. [1]

(ii) The size of this acceleration is $9.7 \times 10^{-3}\,\mathrm{m/s^2}$ and the mass of Venus is $4.9 \times 10^{24}\,\mathrm{kg}$. Calculate the size of the force acting on the planet.

force = ... [2]

(c) As time passes, the planet Venus moves a considerable distance around the circular orbit. State why no work is done by the force acting on the planet as it moves.

....

....

.... [2]

28. M/J/10/22/Q9

Fig. 9.1 shows a car braking on a road and coming to rest.

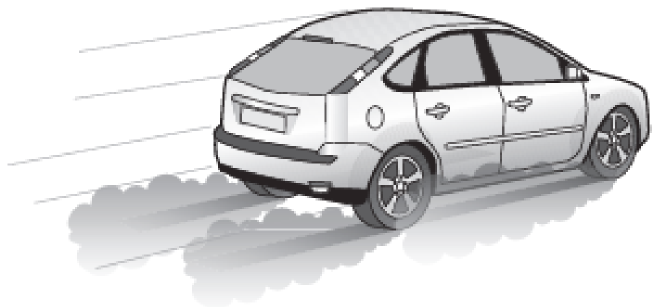


Fig. 9.1

(a) Explain what is meant by

(i) the *thinking distance*,

....

....[1]

(ii) the *braking distance*.

....

....[1]

(b) An engineer conducts a test on the car and finds that the braking distance is greater when the car is fully loaded than when it is unloaded.

(i) Apart from the road conditions, state what must be kept the same in the test.

....[1]

(ii) Explain why the car has a greater braking distance when fully loaded.

....

....[1]

(c) State and explain how one road condition affects the braking distance of the car. Use ideas about friction in your answer.

....

....[2]

(d) Explain how wider tyres affect the pressure of the car on the surface of the road.

....

.... [1]

(e) The car has a total mass of 900 kg and is travelling at 20 m/s. At time t = 0, the driver sees an accident ahead. He applies the brakes at t = 0.60 s to stop the car. After the brakes are applied, the car comes to rest in a further 4.0 s.

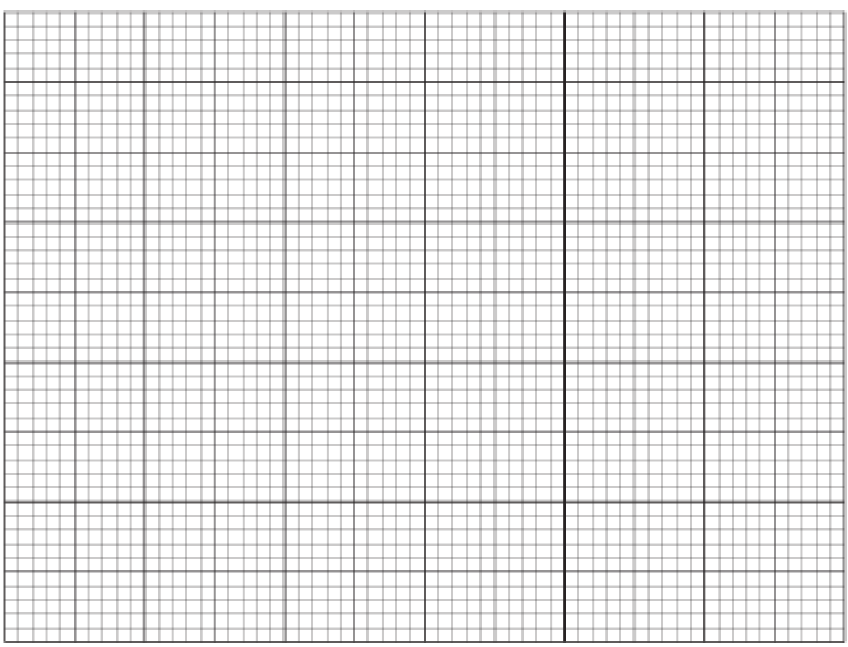
(i) Calculate the deceleration of the car as it brakes.

deceleration = ... [2]

(ii) Calculate the braking force acting on the car.

force = ... [2]

(iii) On Fig. 9.2, draw a speed-time graph for the car as it brakes.



[3]

Fig. 9.2

(iv) State how your graph in **(iii)** can be used to find the total distance travelled by the car.

....

.... [1]